

ABHINAV KAUSHIK

Machine Learning Engineer

New Delhi, India | +91 8700571859 | aabhinav.kaushik@gmail.com | LinkedIn

SUMMARY

Senior Machine Learning Engineer with 8+ years across Healthcare, Telecom and AIOps, owning models end to end from prototype to production. Combines statistical modelling and classical ML for prediction, forecasting, recommendation re-ranking and anomaly detection with LLM applications and AI agents, choosing the right approach for each problem. Works in Python, SQL and PySpark, deploys with FastAPI, Docker and MLflow, and builds dashboards and prototypes with product and business stakeholders. Alongside this, I am building ThinkTree.AI, a non-profit AI learning platform used by NGO teachers to support students with ADHD.

PROFESSIONAL EXPERIENCE

Senior Machine Learning Engineer | United Health Group Bengaluru, India | Nov 2025 - Present

- Deployed a production lab-extraction agent with Claude Opus 4.7 and Amazon Textract, integrated with LIMS, handling about 1,800 lab documents and automating the work behind 800 Jira tickets and PRD tasks.
- Engineered a production multi-agent analytics chatbot in LangGraph for AARP medical plans, answering business questions with PySpark analysis, multi-turn reasoning and live charts, cutting manual analysis by 98%.
- Built automated evaluation and observability pipelines for Question Answering and Summary Agents with LLM evaluators, tracing and failure detection, reducing human-in-the-loop review by 43%.
- Fine-tuned a cross-encoder on positive and negative pairs to re-rank dense retrieval over AARP survey data, improving retrieval accuracy.

Senior Machine Learning Engineer | Viavi Solutions Chennai, India | Sep 2024 - Nov 2025

- Developed an internal RAG-based Project Discovery Assistant in LangGraph combining semantic retrieval with LLM reasoning, reaching a 90% match rate against existing organizational solutions.
- Built an event-correlation engine clustering alarms with Sentence-BERT and DBSCAN to trace the cascading effect of network alarm floods back to the root event, improving semantic alarm classification by 73.5%.
- Turned AIOps and GeoPerformance client pain points into 15+ rapid AI prototypes, with 7 moving into production or paid client work.

Data Scientist | Viavi Solutions Bengaluru, India | Nov 2021 - Sep 2024

- Deployed a production SLA breach prediction model with LightGBM, FastAPI and MLflow, pairing it with GPT-3.5 for real-time root-cause analysis and remediation per breach, reducing downtime by 67%.
- Built anomaly detection and demand forecasting models on 15+ KPIs across 500+ telecom network devices with Isolation Forest and Holt-Winters, helping operations teams detect issues earlier and plan proactively.
- Delivered technical POCs and consulting for 15+ clients across 3+ AIOps and ML platforms, contributing to an 11% increase in POC and mentoring revenue.

Data Analyst | Optum Gurugram, India | Jun 2021 - Nov 2021

- Improved an existing recommendation system with a reranker for individual and child plan recommendations, optimizing renewal suggestions and increasing conversion by 2%.
- Developed a Python and Flask parser that automated extraction and transformation of raw data into structured datasets, feeding downstream analytics and reporting.

Data Engineer | Nippon Data Systems Gurugram, India | Nov 2019 - May 2021

- Engineered SQL and PySpark data pipelines automating BAU medical-audit reporting on large healthcare datasets, cutting manual reporting effort by 95%.
- Wrote complex SQL queries with multi-table joins, CTEs and window functions, delivering ad-hoc reports and analyses for multiple stakeholder teams.
- Built real-time log observability for the COVID QRT team with Elasticsearch and Kibana, improving traceability efficiency by 80%.

Associate Consultant | Saksoft Gurugram, India | Dec 2017 - Nov 2019

- Built an XGBoost propensity model predicting personal-loan applications within 7 days of contact, helping business teams target outreach and prioritize customers.
- Built customer dashboards for financial and product insights with QlikSense, SQL and SAP HANA, increasing user engagement by 40%.

PROJECTS

Think Tree | ML Study Platform

- Built a non-profit AI-powered ML study platform for people with ADHD and context-switching challenges, designed to help users organize learning and maintain study context.

Freight Price Estimation Model

- Built a price estimation and capacity forecasting model for road freight lanes, combining LightGBM regression on lane, distance and seasonality features with Holt-Winters baselines.

- Validated it with backtesting and hypothesis tests against rule-based pricing, served estimates through FastAPI with MLflow tracking, and shared results in a Streamlit dashboard.
- Technologies:** Python, SQL, LightGBM, Streamlit, Jupyter, FastAPI, MLflow, Pandas

RFP Multi-Agent Proposal Automation Platform

- Built a multi-agent RFP platform where an LLM supervisor orchestrates intent detection, retrieval, proposal generation, document matching and market research, with RAG over historical proposals.

Financial Fraud Detection | Mar 2025

- Built a fraud detection and reasoning workflow on the JPMorgan Synthetic Dataset, pairing LightGBM scoring with LLM-generated explanations for interpretable financial-risk analysis.

Restaurant Feedback Analyzer | Aspect-Based Sentiment Analysis

- Built an NLP pipeline to extract review aspects and determine sentiment at the aspect level, enabling granular insights into customer experience.

EDUCATION

Master of Science, Machine Learning and AI Liverpool John Moores University	Jan 2021 - Jan 2023
Bachelor of Technology, Information Technology Jamia Hamdard	Aug 2013 - Aug 2017

TECHNICAL SKILLS

AI / GenAI: Generative AI, LLM, RAG, Agentic RAG, Agentic AI, LangGraph, LangChain, Claude, OpenAI, Embeddings, Cross-Encoder, Fine-Tuning, Hugging Face Transformers, NLP, Tool Orchestration, Amazon Textract
Machine Learning: PyTorch, Scikit-learn, XGBoost, LightGBM, Random Forest, Isolation Forest, LSTM, SBERT, DBSCAN, Time Series Forecasting, Demand Forecasting, Anomaly Detection, Statistical Modeling, Regression, Deep Learning
Evals / Observability: LLM Evals, LLM Evaluators, Agent Tracing, Observability, Elasticsearch, Kibana, Log Analysis
Programming / Data: Python, SQL, PySpark, Pandas, NumPy, FastAPI, Flask, Hive, Spark SQL, REST API, Pydantic, Unix
MLOps / Cloud: MLflow, Docker, Kubernetes, Airflow, DVC, Terraform, Jenkins, Bamboo, Bitbucket, GCP, Azure, Vercel
Data Platforms: Databricks, Databricks SQL, Apache Spark, Supabase
Databases: TimescaleDB, MySQL, Vector DB, pgvector, ArangoDB, Cosmos DB
Search: Apache Solr, Lucene Indexing, Information Retrieval
BI: Power BI, QlikSense, SAP HANA

LANGUAGES

English: Native

ADDITIONAL INFORMATION

- Network Intern, Airport Authority of India
- Hadoop Intern, Ducat Technologies